

Impact Map Explainer

Reliability-Qualified Nighttime Lights a proxy for functional recovery

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This impact map traces how the research moves from a practical disaster-response problem to public benefit. Its five elements, namely, the problem, solution, outputs, beneficiaries and impact, show how satellite information could complement field evidence and support more informed recovery decisions.

1. The problem

After a major disaster, damaged buildings, blocked roads, and collapsed bridges can isolate communities. Ground surveys remain essential, but access may be dangerous, slow or impossible. Decision-makers may therefore struggle to see where disruption is greatest, which places are recovering, and which communities still need help. Remote and underserved areas are especially likely to be missed.

2. The solution

The proposed solution uses satellite observations of nighttime lights: light produced by streets, homes, businesses, and infrastructure after dark. A sharp reduction may indicate disruption, while returning light may suggest that activity and services are resuming.

Nighttime lights are only a proxy for functional recovery, not a direct measure of power restoration, building repair, or wellbeing. Clouds can also hide the ground. This reliability-qualified approach therefore checks whether the satellite obtained a sufficiently clear and trustworthy view before interpreting changes in brightness. The results are then compared with local knowledge, field reports, and infrastructure data.

3. Outputs: what we show

The research will produce free and accessible daily recovery maps when reliable observations are available, detailed time-series profiles, and maps of recurring vulnerability, and resilience patterns. Quality warnings will show when clouds or missing observations limit interpretation. Satellite evidence will be presented alongside ground information rather than as a replacement for it.

4. Beneficiaries: who it helps

At the local level, underserved communities may gain visibility when formal reporting is incomplete, while local leaders can identify places needing continued attention. Power crews could use the maps to help prioritise inspections and repairs. Emergency

teams could locate isolated settlements, while aid groups could compare recovery levels when allocating assistance.

National disaster and energy agencies could use wider regional patterns to coordinate resources. International humanitarian organisations and researchers may also use the accessible evidence and reproducible approach. Although developed through a Philippine disaster case, the approach could be transferred to other cloud-prone and data-limited regions.

5. Impact: what changes

The intended instrumental impacts are faster assessments, more targeted repairs, fairer aid, wider coverage of remote areas, and clearer monitoring of recovery. These changes could improve public safety and help restore essential services and livelihoods sooner.

Socially, the research may reduce the chance that less visible communities are overlooked. Economically, it could help limited staff, funding, and supplies reach areas with the greatest continuing need while reducing duplicated fieldwork.

The conceptual impact also promotes a more responsible way of interpreting satellite data. Recovery is not assumed to be smooth or visible every day, and “not observed” is not treated as “not recovered.” Because these maps may influence priorities, agencies and communities have an interest in how places are represented. Oversimplified comparisons could stigmatise communities or misdirect assistance, so uncertainty and local consultation are essential.

The contribution beyond academia is therefore not simply another satellite map. It is a cautious and practical approach for turning freely available observations into more equitable, transparent, and accountable disaster-recovery decisions.